PRAKTIKUM BASIS DATA

MODUL 3

(Scripting SQL Basic Query – Pemrosesan Satu Tabel)

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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TP

1. **Soal 1**

Worksheet | Query Builder

```
CREATE TABLE mahasiswa (
    id NUMBER PRIMARY KEY,
    nama VARCHAR2(50),
    tempat_lahir VARCHAR2(50),
    tanggal_lahir DATE,
    no_hp VARCHAR2(15),
    email VARCHAR2(50),
    tinggi_badan NUMBER,
    berat_badan NUMBER
);
```

output:

Columns | Data | Model | Constraints | Grants | Statistics | Triggers | Flashback | Dependencies | Details | Partitions | Indexe

Actions...

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	ID	NUMBER	No	(null)	1	(null)
2	NAMA	VARCHAR2(50 BYTE)	Yes	(null)	2	(null)
3	TEMPAT_LAHIR	VARCHAR2(50 BYTE)	Yes	(null)	3	(null)
4	TANGGAL_LAHIR	DATE	Yes	(null)	4	(null)
5	NO_HP	VARCHAR2(15 BYTE)	Yes	(null)	5	(null)
6	EMAIL	VARCHAR2(50 BYTE)	Yes	(null)	6	(null)
7	TINGGI_BADAN	NUMBER	Yes	(null)	7	(null)
8	BERAT_BADAN	NUMBER	Yes	(null)	8	(null)

2. Soal 2

Worksheet | Query Builder

```
INSERT INTO mahasiswa VALUES (1,'Ghina','Semarang',DATE '2000-05-10','08123456789','ghina@gmail.com',165,55);
INSERT INTO mahasiswa VALUES (2,'Nana','Bandung',DATE '1999-08-11','08234567890','nana@gmail.com',160,50);
INSERT INTO mahasiswa VALUES (3,'Soma','Bogor',DATE '2001-03-15','08345678901','soma@gmail.com',170,65);
INSERT INTO mahasiswa VALUES (4,'Cya','Depok',DATE '2000-07-20','08456789012','cya@gmail.com',158,48);
INSERT INTO mahasiswa VALUES (5,'Rara','Bekasi',DATE '2002-01-12','08567890123','rara@gmail.com',162,52);
INSERT INTO mahasiswa VALUES (6,'Andi','Jakarta',DATE '2000-11-22','08678901234','andi@gmail.com',172,68);
INSERT INTO mahasiswa VALUES (7,'Budi','Semarang',DATE '1999-09-10','08789012345','budi@gmail.com',168,70);
INSERT INTO mahasiswa VALUES (8,'Citra','Surabaya',DATE '2001-02-18','08890123456','citra@gmail.com',160,55);
INSERT INTO mahasiswa VALUES (9,'Doni','Yogyakarta',DATE '2000-04-25','08901234567','doni@gmail.com',175,72);
INSERT INTO mahasiswa VALUES (10,'Eka','Medan',DATE '1998-12-05','08112233445','eka@gmail.com',165,60);
INSERT INTO mahasiswa VALUES (11,'Fajar','Makassar',DATE '2001-06-30','08223344556','fajar@gmail.com',178,75);
INSERT INTO mahasiswa VALUES (12,'Gita','Padang',DATE '2002-08-03','08334455667','gita@gmail.com',163,54);
INSERT INTO mahasiswa VALUES (13,'Hadi','Malang',DATE '2000-10-10','08445566778','hadi@gmail.com',170,69);
INSERT INTO mahasiswa VALUES (14,'Intan','Solo',DATE '2001-05-17','08556677889','intan@gmail.com',159,51);
INSERT INTO mahasiswa VALUES (15,'Joko','Bali',DATE '1999-01-27','08667788990','joko@gmail.com',173,74);
```

output:

Columns | Data | Model | Constraints | Grants | Statistics | Triggers | Flashback | Dependencies | Details | Partitions | Indexes | SQL

Sort.. | Filter:

	ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NO_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN
1	1	Ghina	Semarang	10-05-2000	08123456789	ghina@gmail.com	165	55
2	2	Nana	Bandung	11-08-1999	08234567890	nana@gmail.com	160	50
3	3	Soma	Bogor	15-03-2001	08345678901	soma@gmail.com	170	65
4	4	Cya	Depok	20-07-2000	08456789012	cya@gmail.com	158	48
5	5	Rara	Bekasi	12-01-2002	08567890123	rara@gmail.com	162	52
6	6	Andi	Jakarta	22-11-2000	08678901234	andi@gmail.com	172	68
7	7	Budi	Semarang	10-09-1999	08789012345	budi@gmail.com	168	70
8	8	Citra	Surabaya	18-02-2001	08890123456	citra@gmail.com	160	55
9	9	Doni	Yogyakarta	25-04-2000	08901234567	doni@gmail.com	175	72
10	10	Eka	Medan	05-12-1998	08112233445	eka@gmail.com	165	60
11	11	Fajar	Makassar	30-06-2001	08223344556	fajar@gmail.com	178	75
12	12	Gita	Padang	03-08-2002	08334455667	gita@gmail.com	163	54
13	13	Hadi	Malang	10-10-2000	08445566778	hadi@gmail.com	170	69
14	14	Intan	Solo	17-05-2001	08556677889	intan@gmail.com	159	51
15	15	Joko	Bali	27-01-1999	08667788990	joko@gmail.com	173	74

3. Soal 3

Worksheet | Query Builder

```sql
SELECT
LOWER(nama) AS lower_nama,
UPPER(nama) AS upper_nama,
INITCAP(nama) AS initcap_nama,
CONCAT(nama,email) AS gabung,
LENGTH(nama) AS panjang_nama,
SUBSTR(nama,1,3) AS tiga_huruf,
LPAD(nama,10,'*') AS lpad_nama,
RPAD(nama,10,'*') AS rpad_nama,
INSTR(email,'@') AS posisi_at,
REPLACE(nama,'a','A') AS replace_nama
FROM mahasiswa;
```

output:

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3 | Query Result 4 | Query Result 5 | Que...

SQL | All Rows Fetched: 15 in 0.043 seconds

	LOWER_NAMA	UPPER_NAMA	INITCAP_NAMA	GABUNG	PANJANG_NAMA	TIGA_HURUF	LPAD_NAMA	RPAD_NAMA	POSISI_AT	REPLACE_NAMA
1	ghina	GHINA	Ghina	Ghinaghina@gmail.com	5	Ghi	*****Ghina	Ghina*****	6	GhinA
2	nana	NANA	Nana	Nananana@gmail.com	4	Nan	******Nana	Nana******	5	NAnA
3	soma	SOMA	Soma	Somasoma@gmail.com	4	Som	******Soma	Soma******	5	SomA
4	cya	CYA	Cya	Cyacya@gmail.com	3	Cya	*******Cya	Cya*******	4	CyA
5	rara	RARA	Rara	Rararara@gmail.com	4	Rar	******Rara	Rara******	5	RArA
6	andi	ANDI	Andi	Andiandi@gmail.com	4	And	******Andi	Andi******	5	Andi
7	budi	BUDI	Budi	Budibudi@gmail.com	4	Bud	******Budi	Budi******	5	Budi
8	citra	CITRA	Citra	Citracitra@gmail.com	5	Cit	*****Citra	Citra*****	6	CitrA
9	doni	DONI	Doni	Donidoni@gmail.com	4	Don	******Doni	Doni******	5	Doni
10	eka	EKA	Eka	Ekaeka@gmail.com	3	Eka	*******Eka	Eka*******	4	EkA
11	fajar	FAJAR	Fajar	Fajarfajar@gmail.com	5	Faj	*****Fajar	Fajar*****	6	FAjAr
12	gita	GITA	Gita	Gitagita@gmail.com	4	Git	******Gita	Gita******	5	GitA
13	hadi	HADI	Hadi	Hadihadi@gmail.com	4	Had	******Hadi	Hadi******	5	HAdi
14	intan	INTAN	Intan	Intanintan@gmail.com	5	Int	*****Intan	Intan*****	6	IntAn
15	joko	JOKO	Joko	Jokojoko@gmail.com	4	Jok	******Joko	Joko******	5	Joko

4. Soal 4

```sql
SELECT
TO_CHAR(tanggal_lahir,'MM/MON/MONTH') AS format_bulan,
TO_CHAR(tanggal_lahir,'DD/DY') AS format_hari,
TO_CHAR(tanggal_lahir,'YYYY/YY') AS format_tahun,
TO_CHAR(SYSDATE,'HH24:MI:SS') AS jam_sekarang,
ADD_MONTHS(tanggal_lahir,6) AS tambah_6_bulan,
SYSDATE AS tanggal_sekarang,
LAST_DAY(tanggal_lahir) AS akhir_bulan_lahir,
NEXT_DAY(tanggal_lahir,'MONDAY') AS hari_berikutnya,
MONTHS_BETWEEN(SYSDATE,tanggal_lahir) AS usia_dalam_bulan,
ROUND(tanggal_lahir,'YEAR') AS pembulatan_tahun
FROM mahasiswa
WHERE nama = 'Fikri';
```

output:

Query Result

SQL | All Rows Fetched: 15 in 0.004 seconds

	BULAN_ANGKA	BULAN_SINGKAT	BULAN_LENGKAP	HARI_ANGKA	HARI_SINGKAT	TAHUN_PANJANG	TAHUN_PENDEK	SELISIH_BULAN
1	05	MAY	MAY	10	WED	2000	00	310
2	08	AUG	AUGUST	11	WED	1999	99	318,98302904719235364396654719235364396 7
3	03	MAR	MARCH	15	THU	2001	01	299,85399678912783751493428912783751493 4
4	07	JUL	JULY	20	THU	2000	00	307,69270646654719235364396654719235364 4
5	01	JAN	JANUARY	12	SAT	2002	02	289,95077098267622461170848267622461170 8
6	11	NOV	NOVEMBER	22	WED	2000	00	303,62819033751493428912783751493428912 8
7	09	SEP	SEPTEMBER	10	FRI	1999	99	318
8	02	FEB	FEBRUARY	18	SUN	2001	01	300,75722259557945041816009557945041816
9	04	APR	APRIL	25	TUE	2000	00	310,53141614396654719235364396654719235 4
10	12	DEC	DECEMBER	05	SAT	1998	98	327,17657743428912783751493428912783751 5
11	06	JUN	JUNE	30	SAT	2001	01	296,37012582138590203106332138590203106 3
12	08	AUG	AUGUST	03	SAT	2002	02	283,24109356332138590203106332138590203 1
13	10	OCT	OCTOBER	10	TUE	2000	00	305
14	05	MAY	MAY	17	THU	2001	01	297,78948066009557945041816009557945041 8
15	01	JAN	JANUARY	27	WED	1999	99	325,46690001493428912783751493428912783 8

	TO_DATE('10-05-2000','DD-MM-YYYY')
1	10-05-2000

5. Soal 5

Worksheet | Query Builder

```
SELECT
ABS(-10) AS hasil_abs,
MOD(10,3) AS hasil_mod,
FLOOR(10.8) AS hasil_floor,
POWER(2,3) AS hasil_power,
ROUND(12.345,2) AS hasil_round,
CEIL(10.2) AS hasil_ceil
FROM dual;
```

output:

Query Result

All Rows Fetched: 1 in 0.01 seconds

	HASIL_ABS	HASIL_MOD	HASIL_FLOOR	HASIL_POWER	HASIL_ROUND	HASIL_CEIL
1	10	1	10	8	12,35	11

Query Result

All Rows Fetched: 15 in 0.162 seconds

	ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NO_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN	HASIL_ABS	HASIL_MOD	HASIL_FLOOR	HASIL_POWER	HASIL_ROUND	HASIL_CEIL
1	1	Ghina	Semarang	10-05-2000	08123456789	ghina@gmail.com	165	55	10	1	10	8	12,35	11
2	2	Nana	Bandung	11-08-1999	08234567890	nana@gmail.com	160	50	10	1	10	8	12,35	11
3	3	Soma	Bogor	15-03-2001	08345678901	soma@gmail.com	170	65	10	1	10	8	12,35	11
4	4	Cya	Depok	20-07-2000	08456789012	cya@gmail.com	158	48	10	1	10	8	12,35	11
5	5	Rara	Bekasi	12-01-2002	08567890123	rara@gmail.com	162	52	10	1	10	8	12,35	11
6	6	Andi	Jakarta	22-11-2000	08678901234	andi@gmail.com	172	68	10	1	10	8	12,35	11
7	7	Budi	Semarang	10-09-1999	08789012345	budi@gmail.com	168	70	10	1	10	8	12,35	11
8	8	Citra	Surabaya	18-02-2001	08890123456	citra@gmail.com	160	55	10	1	10	8	12,35	11
9	9	Doni	Yogyakarta	25-04-2000	08901234567	doni@gmail.com	175	72	10	1	10	8	12,35	11
10	10	Eka	Medan	05-12-1998	08112233445	eka@gmail.com	165	60	10	1	10	8	12,35	11
11	11	Fajar	Makassar	30-06-2001	08223344556	fajar@gmail.com	178	75	10	1	10	8	12,35	11
12	12	Gita	Padang	03-08-2002	08334455667	gita@gmail.com	163	54	10	1	10	8	12,35	11
13	13	Hadi	Malang	10-10-2000	08445566778	hadi@gmail.com	170	69	10	1	10	8	12,35	11
14	14	Intan	Solo	17-05-2001	08556677889	intan@gmail.com	159	51	10	1	10	8	12,35	11
15	15	Joko	Bali	27-01-1999	08667788990	joko@gmail.com	173	74	10	1	10	8	12,35	11